

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I Atmakuri Sai Mrudula (192372087) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. Sai Mrudula

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge

servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

E-commerce companies experience sudden increases in customer traffic during seasonal sales such as Black Friday, Diwali, Christmas, or New Year offers. During these periods, thousands or even millions of users access the website simultaneously to purchase products. If the infrastructure cannot handle the increased demand, customers may experience slow response times, website crashes, and failed transactions. Cloud computing solves this challenge through Auto Scaling, Load Balancing, and Cloud Monitoring, which automatically manage resources according to workload demands.

Auto Scaling

Auto Scaling automatically adjusts the number of application servers based on workload.

When CPU utilization, memory usage, or network traffic increases beyond a predefined threshold, additional virtual machines or containers are automatically launched. During periods of low demand, unnecessary resources are removed.

Benefits include:

- Automatic resource allocation
- Improved application performance
- Reduced operational cost
- Better scalability
- Efficient resource utilization

Load Balancing

Load balancing distributes incoming user requests evenly among multiple servers.

Instead of directing all users to one server, the load balancer sends each request to the least busy healthy server. If one server fails, traffic is automatically redirected to another available server.

Benefits include:

- Prevents server overload
- Improves response time
- Increases application availability
- Provides fault tolerance
- Reduces downtime

Reducing Downtime and Improving User Experience

The combination of Auto Scaling and Load Balancing ensures uninterrupted service.

When customer traffic suddenly increases during sales, Auto Scaling launches additional servers within minutes. The Load Balancer immediately starts distributing incoming requests to these new servers.

Customers experience:

- Faster page loading
- Smooth checkout process
- Reduced waiting time
- Fewer application crashes

- Better shopping experience

Even if one application server fails unexpectedly, users are automatically redirected to healthy servers without noticing any interruption.

Role of Monitoring Tools

Monitoring tools continuously observe system performance.

They monitor:

- CPU utilization
- Memory usage
- Network traffic
- Application response time
- Disk utilization
- Error rates
- Database performance

If abnormal behavior is detected, alerts are immediately sent to administrators. Monitoring services also trigger Auto Scaling policies automatically.

Examples include:

- Amazon CloudWatch
- Azure Monitor
- Google Cloud Monitoring

Real-World Example

Consider an e-commerce company such as **Amazon** during the Black Friday sale.

Millions of customers visit the website simultaneously. Cloud monitoring detects increased traffic and automatically launches additional application servers. The Load Balancer distributes customer requests among these servers, preventing overload. When the sale ends, unused servers are automatically terminated, reducing cloud costs.

As a result, customers experience fast website performance, successful payments, and uninterrupted shopping.

Advantages

- Handles unpredictable traffic efficiently.
- Provides automatic scalability.
- Minimizes application downtime.
- Improves customer satisfaction.
- Optimizes cloud costs.
- Ensures high availability.
- Supports business continuity.
- Provides better resource utilization.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Organizations generate large amounts of important data, including customer information, financial records, business documents, application databases, and multimedia files. Losing this data due to accidental deletion, hardware failure, cyberattacks, or natural disasters can severely affect business operations. Cloud storage provides a reliable and cost-effective solution for securely storing backup data while ensuring high durability, redundancy, and availability.

Cloud storage allows organizations to store data in remote cloud data centers instead of maintaining physical storage devices within their own premises.

Cloud providers automatically manage storage infrastructure, backup processes, hardware maintenance, and scalability. Users can increase storage capacity whenever required without purchasing additional hardware.

Cloud Storage Durability and Data Redundancy

Cloud providers maintain multiple copies of every stored file.

Data is replicated across multiple disks, servers, Availability Zones, and sometimes different geographical regions.

If one storage device fails, another copy remains immediately available.

Most cloud providers guarantee durability of 99.999999999% by using:

- Multiple storage replicas
- Automatic data replication
- RAID technologies
- Cross-region replication
- Regular integrity verification

This ensures that backup data remains available even during hardware failures or disasters.

Encryption and Access Control

Security is extremely important for backup data.

Cloud providers use AES-256 encryption to protect stored data.

When data travels between users and cloud storage, TLS encryption secures communication.

Access control is managed through Identity and Access Management (IAM).

Organizations implement:

- Role-Based Access Control (RBAC)
- Multi-Factor Authentication (MFA)
- Least Privilege Access
- Secure API Keys
- Audit Logging

These security measures prevent unauthorized users from accessing sensitive backup data.

Cost Advantages

Traditional storage systems require organizations to purchase expensive storage servers, networking equipment, backup devices, cooling systems, and physical infrastructure.

Cloud storage eliminates these expenses.

Organizations pay only for the storage they actually use.

Additional cost advantages include:

- No hardware maintenance.
- Automatic scalability.
- Lower electricity costs.
- No backup hardware purchases.
- Reduced administrative effort.
- Lifecycle storage policies.
- Low-cost archival storage.

Overall, cloud storage significantly reduces Total Cost of Ownership (TCO).

Disaster Recovery Use Case

Consider a hospital storing patient medical records.

A ransomware attack encrypts the hospital's primary database.

Since backup data is securely stored in cloud storage with versioning and cross-region replication, administrators immediately restore the latest clean backup.

The hospital resumes operations within a short time without losing important patient records.

This demonstrates how cloud storage supports effective disaster recovery.

Advantages

- High durability.
- Secure data storage.
- Automatic redundancy.
- Low operational cost.
- Fast disaster recovery.
- Easy scalability.
- Global accessibility.
- Reliable backup management.

Cloud storage provides secure, scalable, and highly durable backup solutions for modern organizations. Through automatic replication, encryption, access control, and disaster recovery capabilities, cloud storage protects valuable business information while reducing infrastructure costs and ensuring continuous business operations.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Mobile applications today serve users from different countries and continents. Users expect applications to load quickly regardless of their geographical location. If application data is served only from one central server, users located far away may experience high latency and slow response times. Cloud providers solve this challenge using Content Delivery Networks (CDNs) and globally distributed Edge Servers.

A Content Delivery Network (CDN) is a network of geographically distributed servers that store cached copies of application content.

Instead of downloading content from the origin server every time, users receive data from the nearest edge server.

This significantly reduces latency and improves application performance.

How CDNs Improve Performance

Popular images, videos, application files, software updates, and web pages are stored on edge servers.

When users request content, the CDN automatically delivers it from the closest server.

Benefits include:

- Faster loading time.
- Reduced network latency.
- Lower bandwidth consumption.
- Reduced load on origin servers.
- Better application responsiveness.

Role of Edge Servers

Edge servers are cloud servers located near users.

Instead of travelling thousands of kilometers to reach the main server, requests are handled locally.

Advantages include:

- Faster response.
- Reduced buffering.
- Better streaming quality.
- Improved gaming experience.
- Lower packet loss.

Global Availability

Cloud providers maintain data centers worldwide.

Services include:

- Global Load Balancers.
- Multiple Regions.
- Availability Zones.
- Cross-region replication.
- Auto Scaling.

Users are automatically connected to the nearest healthy region.

Real-Time Streaming Example

Netflix stores video content on CDN edge servers worldwide.

When users watch movies, videos are delivered from nearby edge locations instead of the central server.

This reduces buffering and provides high-quality streaming even during peak traffic.

Advantages

- Low latency.
- Faster downloads.
- Improved user experience.
- Reduced server load.
- Better global availability.
- High scalability.
- Reliable content delivery.

CDNs and Edge Servers significantly improve the performance of globally deployed mobile applications. By delivering content from locations closer to users, cloud providers reduce latency, improve responsiveness, and ensure reliable application performance worldwide.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Financial institutions such as banks, insurance companies, and investment firms manage highly sensitive information, including customer accounts, transaction records, credit card details, and personal financial data. Any unauthorized access, data breach, or regulatory violation can result in financial losses, legal penalties, and loss of customer trust. Therefore, cloud computing environments used by financial institutions must provide strong security, regulatory compliance, continuous monitoring, and data protection. Modern cloud providers offer various security services that help organizations comply with industry regulations while maintaining high availability and performance.

Cloud Regulatory Compliance and Auditing

Cloud providers support compliance by offering infrastructure that meets international security standards.

Common compliance standards include:

- PCI DSS (Payment Card Industry Data Security Standard)
- ISO 27001 (Information Security Management)
- SOC 1 and SOC 2 (System and Organization Controls)
- GDPR (General Data Protection Regulation)
- HIPAA (Health Insurance Portability and Accountability Act)
- RBI Guidelines (for financial institutions in India)

Cloud providers regularly perform security audits, vulnerability assessments, penetration testing, and compliance reporting to ensure their infrastructure meets these standards. Organizations can also generate audit reports to demonstrate compliance during inspections or regulatory reviews.

Role of Logging

Logging records every activity performed within the cloud environment.

Logs capture information such as:

- User login attempts
- Financial transactions
- File access
- Database queries
- Administrative actions
- API requests
- Security events
- Failed authentication attempts

These logs help administrators investigate incidents, detect suspicious activities, and perform forensic analysis after security breaches.

Logs also provide evidence required during regulatory audits.

Role of Monitoring

Monitoring continuously observes cloud resources and application performance.

Monitoring tools collect information about:

- CPU utilization
- Memory usage
- Network traffic
- Application response time
- Database performance
- Unauthorized access attempts
- Security threats
- Service availability

Whenever abnormal activity is detected, automated alerts notify administrators immediately.

Examples include:

- Amazon CloudWatch
- Azure Monitor
- Google Cloud Operations Suite

Continuous monitoring helps organizations detect security threats before they cause serious damage.

Role of Encryption

Encryption converts readable information into an unreadable format that can only be accessed using authorized encryption keys. Cloud providers implement encryption in two ways.

Encryption at Rest

Data stored inside cloud databases, storage systems, and backups is encrypted using AES-256 encryption. Even if storage devices are stolen, attackers cannot read encrypted data.

Encryption in Transit

Data travelling between users and cloud servers is protected using TLS (Transport Layer Security) encryption. This prevents attackers from intercepting confidential financial information during transmission. Cloud providers also offer Key Management Services that securely manage encryption keys.

Ensuring Data Sovereignty

Data sovereignty means that data remains stored and processed within a specific country or geographic region according to local laws.

Financial institutions can satisfy data sovereignty requirements by:

- Choosing cloud regions located within the required country.
- Restricting cross-border data transfers.
- Using region-specific storage services.
- Configuring backup locations within approved regions.
- Applying country-specific compliance policies.

This ensures customer financial information remains under national legal jurisdiction.

Example of Sensitive Financial Transactions

Consider an online banking application. A customer transfers money from one account to another. The login request is authenticated using Multi-Factor Authentication (MFA). Transaction details are encrypted before transmission. The payment request is processed inside a secure cloud environment. Every activity is recorded in audit logs. Monitoring services continuously analyze the transaction for suspicious behavior such as unusual login locations or abnormal transfer amounts.

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Modern software applications are expected to be developed, tested, and deployed rapidly while maintaining high performance and reliability. Traditional deployment methods using physical servers or virtual machines often require significant resources and time. Containerization has emerged as a modern solution that packages applications together with all required libraries, dependencies, and runtime environments. Containers provide faster deployment, better resource utilization, and consistent execution across different cloud environments. To manage large numbers of containers efficiently, organizations use container orchestration platforms such as Kubernetes.

Containers vs Virtual Machines

Containers and Virtual Machines both provide isolated environments for applications, but they differ significantly.

Containers

Containers share the host operating system kernel. They start within seconds and require very little memory and CPU resources. Because of their lightweight design, hundreds of containers can run on a single server. Containers are ideal for cloud-native applications and microservices.

Virtual Machines

Virtual Machines include a complete guest operating system. Each VM requires dedicated CPU, memory, and storage resources. They consume more hardware resources and require longer startup times. Virtual Machines are more suitable when complete operating system isolation is required. Overall, containers provide better performance, higher density, faster deployment, and lower resource consumption.

Container Orchestration Using Kubernetes

Managing hundreds or thousands of containers manually is extremely difficult. Kubernetes automates container deployment, scaling, networking, monitoring, and recovery. Its major responsibilities include:

- Automatic deployment
- Container scheduling
- Load balancing
- Auto Scaling
- Self-healing
- Rolling updates
- Automatic rollback

- Service discovery
- Resource management

Deployment and Scaling

- Kubernetes simplifies application deployment through automation.
- Applications are deployed as Pods, which contain one or more containers.
- Horizontal Pod Autoscaler automatically creates additional container instances whenever CPU or memory utilization increases.
- Cluster Autoscaler automatically adds new worker nodes if existing servers cannot accommodate more containers.
- When workload decreases, unnecessary containers and worker nodes are removed automatically, reducing cloud costs.

Portability Across Cloud Environments

One of the greatest advantages of containers is portability. Since containers include all required dependencies, the same application can run without modification on:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)
- Private Cloud
- Hybrid Cloud
- Local Data Centers

This eliminates compatibility issues between different operating systems and cloud providers. Organizations can migrate workloads easily without rewriting applications.

Microservices-Based Application Example

Consider an online food delivery application. Instead of developing one large application, the system is divided into independent microservices such as:

- User Authentication Service
- Restaurant Service
- Food Ordering Service
- Payment Service
- Delivery Tracking Service
- Notification Service